PGCP-AC AUGUST 26
Assignment No- 2

1. Write a SQL query to create Employee & Department table with specific columns and constraints.(**Note**- Create Table, insert data as per your Choice)

```
CREATE DATABASE dbt_assignment;
USE dbt_assignment;
CREATE TABLE Department (
   DeptID INT PRIMARY KEY,
   DeptName VARCHAR(50),
   City VARCHAR(50)
);
CREATE TABLE Employee (
   EmpID INT PRIMARY KEY,
   EmpName VARCHAR(50),
   Salary DECIMAL(10,2),
   JoiningDate DATE,
   DeptID INT,
   FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
);
INSERT INTO Department VALUES
(10, 'IT', 'Pune'),
(20, 'HR', 'Mumbai'),
(30, 'Sales', 'Nagpur');
INSERT INTO Employee VALUES
(101, 'Amit', 40000, '2024-01-10', 10),
(102, 'Priya', 55000, '2023-05-15', 20),
(103, 'Rahul', 60000, '2022-08-20', 10),
(104, 'Neha', 45000, '2024-03-12', 30),
(105, 'Mansi', 70000, '2021-11-05', 20);
SELECT * FROM Department;
```

```
mysql> CREATE DATABASE dbt_assignment;
Query OK, 1 row affected (0.06 sec)

mysql> USE dbt_assignment;
Database changed
mysql> CREATE TABLE Department (
    ->     DeptID INT PRIMARY KEY,
    ->     DeptName VARCHAR(50),
    ->     City VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Employee (
    ->     EmpID INT PRIMARY KEY,
    ->     EmpName VARCHAR(50),
    ->     Salary DECIMAL(10,2),
    ->     JoiningDate DATE,
    ->     DeptID INT,
    ->     FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Department VALUES
    -> (10, 'IT', 'Pune'),
    -> (20, 'HR', 'Mumbai'),
    -> (30, 'Sales', 'Nagpur');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Employee VALUES
    -> (101, 'Mansi', 40000, '2024-01-10', 10),
    -> (102, 'Snehal', 55000, '2023-05-15', 20),
    -> (103, 'Jayesh', 60000, '2022-08-20', 10),
    -> (104, 'Khushi', 45000, '2024-03-12', 30),
    -> (105, 'Raj', 70000, '2021-11-05', 20);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Employee;
+-------+---------+----------+-------------+--------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID |
+-------+---------+----------+-------------+--------+
|   101 | Mansi   | 40000.00 | 2024-01-10  |     10 |
|   102 | Snehal  | 55000.00 | 2023-05-15  |     20 |
|   103 | Jayesh  | 60000.00 | 2022-08-20  |     10 |
|   104 | Khushi  | 45000.00 | 2024-03-12  |     30 |
|   105 | Raj     | 70000.00 | 2021-11-05  |     20 |
+-------+---------+----------+-------------+--------+
5 rows in set (0.01 sec)
```

2. Write a SQL query to add any new column to an existing table.

ALTER TABLE Employee
ADD Email VARCHAR(100);
USE dbt_assignment;
DESC Employee;

```
mysql> ALTER TABLE Employee
    -> ADD Email VARCHAR(100);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> USE dbt_assignment;
Database changed
mysql> DESC Employee;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| EmpID       | int          | NO   | PRI | NULL    |       |
| EmpName     | varchar(50)  | YES  |     | NULL    |       |
| Salary      | decimal(10,2)| YES  |     | NULL    |       |
| JoiningDate | date         | YES  |     | NULL    |       |
| DeptID      | int          | YES  | MUL | NULL    |       |
| Email       | varchar(100) | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
6 rows in set (0.06 sec)
```

3. Write a SQL query to insert multiple records (At least 5) into a table in a single operation.

```
CREATE TABLE Students (
  StudentID INT PRIMARY KEY,
  StudentName VARCHAR(50),
  City VARCHAR(50)
);
INSERT INTO Students VALUES
(1, 'Mansi', 'Pune'),
(2, 'Snehal', 'Mumbai'),
(3, 'Jayesh', 'Nagpur'),
(4, 'Khushi', 'Nashik'),
(5, 'Raj', 'Shirpur');
SELECT * FROM Students;
```

```
mysql> CREATE TABLE Students (
    ->     StudentID INT PRIMARY KEY,
    ->     StudentName VARCHAR(50),
    ->     City VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Students VALUES
    -> (1, 'Mansi', 'Pune'),
    -> (2, 'Snehal', 'Mumbai'),
    -> (3, 'Jayesh', 'Nagpur'),
    -> (4, 'Khushi', 'Nashik'),
    -> (5, 'Raj', 'Shirpur');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Students;
+-----------+-------------+---------+
| StudentID | StudentName | City    |
+-----------+-------------+---------+
|         1 | Mansi       | Pune    |
|         2 | Snehal      | Mumbai  |
|         3 | Jayesh      | Nagpur  |
|         4 | Khushi      | Nashik  |
|         5 | Raj         | Shirpur |
+-----------+-------------+---------+
5 rows in set (0.00 sec)
```

4. Write a SQL query to delete all records from a table.
   DELETE FROM Students;
   SELECT * FROM Students;

```
mysql> DELETE FROM Students;
Query OK, 5 rows affected (0.01 sec)

mysql> SELECT * FROM Students;
Empty set (0.00 sec)
```

5. Write a SQL query to update salary records from a table where id = 5.
   UPDATE Employee
   SET Salary = 50000
   WHERE EmpID = 5;
   SELECT * FROM Employee
   WHERE EmpID = 5;
   UPDATE Employee
   SET Salary = 50000

WHERE EmpID = 105;
SELECT * FROM Employee
WHERE EmpID = 105;

```
mysql> DELETE FROM Students;
Query OK, 5 rows affected (0.01 sec)

mysql> SELECT * FROM Students;
Empty set (0.00 sec)

mysql> UPDATE Employee
    -> SET Salary = 50000
    -> WHERE EmpID = 5;
Query OK, 0 rows affected (0.02 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT * FROM Employee
    -> WHERE EmpID = 5;
Empty set (0.01 sec)

mysql> UPDATE Employee
    -> SET Salary = 50000
    -> WHERE EmpID = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Employee
    -> WHERE EmpID = 105;
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   105 | Raj     | 50000.00 | 2021-11-05  |     20 | NULL  |
+-------+---------+----------+-------------+--------+-------+
1 row in set (0.00 sec)
```

6. Write a query to grant SELECT privilege on the Emp table to user readonly_user.
   USE dbt_assignment;
   CREATE USER 'readonly_user'@'localhost'
   IDENTIFIED BY 'ReadOnly123';
   GRANT SELECT ON dbt_assignment.Employee
   TO 'readonly_user'@'localhost';
   SHOW GRANTS FOR 'readonly_user'@'localhost';

```
mysql> CREATE USER 'readonly_user'@'localhost'
    -> IDENTIFIED BY 'ReadOnly123';
Query OK, 0 rows affected (0.04 sec)

mysql> GRANT SELECT ON dbt_assignment.Employee
    -> TO 'readonly_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW GRANTS FOR 'readonly_user'@'localhost';
+-----------------------------------------------------------------------------+
| Grants for readonly_user@localhost                                          |
+-----------------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `readonly_user`@`localhost`                           |
| GRANT SELECT ON `dbt_assignment`.`employee` TO `readonly_user`@`localhost` |
+-----------------------------------------------------------------------------+
2 rows in set (0.01 sec)
```

7. Write a query to revoke INSERT privilege on the Emp table from user temp_user.
   CREATE USER 'temp_user'@'localhost'
   IDENTIFIED BY 'Temp123';
   GRANT INSERT ON dbt_assignment.Employee
   TO 'temp_user'@'localhost';
   REVOKE INSERT ON dbt_assignment.Employee
   FROM 'temp_user'@'localhost';
   SHOW GRANTS FOR 'temp_user'@'localhost';

```
mysql> CREATE USER 'temp_user'@'localhost'
    -> IDENTIFIED BY 'Temp123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT INSERT ON dbt_assignment.Employee
    -> TO 'temp_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE INSERT ON dbt_assignment.Employee
    -> FROM 'temp_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'temp_user'@'localhost';
+-----------------------------------------------+
| Grants for temp_user@localhost                |
+-----------------------------------------------+
| GRANT USAGE ON *.* TO `temp_user`@`localhost` |
+-----------------------------------------------+
1 row in set (0.00 sec)
```

8. Write a query to insert a new employee into the Emp table and commit the transaction.
   START TRANSACTION;
   INSERT INTO Employee
   (EmpID, EmpName, Salary, JoiningDate, DeptID)
   VALUES
   (106, 'Snehal', 48000, '2025-01-15', 10);
   COMMIT;

SELECT * FROM Employee
WHERE EmpID = 106;

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO Employee
    -> (EmpID, EmpName, Salary, JoiningDate, DeptID)
    -> VALUES
    -> (106, 'Snehal', 48000, '2025-01-15', 10);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Employee
    -> WHERE EmpID = 106;
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   106 | Snehal  | 48000.00 | 2025-01-15  |     10 | NULL  |
+-------+---------+----------+-------------+--------+-------+
1 row in set (0.00 sec)
```

9. Write a query to update the Salary of an employee with EmpID = 101 and commit the transaction.
   START TRANSACTION;
   UPDATE Employee
   SET Salary = 45000
   WHERE EmpID = 101;
   COMMIT;
   SELECT EmpID, EmpName, Salary
   FROM Employee
   WHERE EmpID = 101;

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO Employee
    -> (EmpID, EmpName, Salary, JoiningDate, DeptID)
    -> VALUES
    -> (106, 'Snehal', 48000, '2025-01-15', 10);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Employee
    -> WHERE EmpID = 106;
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   106 | Snehal  | 48000.00 | 2025-01-15  |     10 | NULL  |
+-------+---------+----------+-------------+--------+-------+
1 row in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

10. Write a query to set a SAVEPOINT before updating the DeptID of an employee.
    START TRANSACTION;
    SAVEPOINT before_update;
    UPDATE Employee
    SET DeptID = 20
    WHERE EmpID = 101;
    SELECT EmpID, EmpName, DeptID
    FROM Employee
    WHERE EmpID = 101;

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT before_update;
Query OK, 0 rows affected (0.01 sec)

mysql> UPDATE Employee
    -> SET DeptID = 20
    -> WHERE EmpID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT EmpID, EmpName, DeptID
    -> FROM Employee
    -> WHERE EmpID = 101;
+-------+---------+--------+
| EmpID | EmpName | DeptID |
+-------+---------+--------+
|   101 | Mansi   |     20 |
+-------+---------+--------+
1 row in set (0.00 sec)
```

11. Write a query to rollback to the previously created SAVEPOINT.
    ROLLBACK TO SAVEPOINT before_update;
    SELECT EmpID, EmpName, DeptID
    FROM Employee
    WHERE EmpID = 101;

```
mysql> ROLLBACK TO SAVEPOINT before_update;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT EmpID, EmpName, DeptID
    -> FROM Employee
    -> WHERE EmpID = 101;
+-------+---------+--------+
| EmpID | EmpName | DeptID |
+-------+---------+--------+
|   101 | Mansi   |     10 |
+-------+---------+--------+
1 row in set (0.00 sec)
```

12. Write a query to truncate all data from the Emp table.
    TRUNCATE TABLE Employee;
    SELECT * FROM Employee;

```
mysql> TRUNCATE TABLE Employee;
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT * FROM Employee;
Empty set (0.00 sec)
```

13. Write a query to list all employees whose salary is greater than 50,000.
    SELECT *
    FROM Employee
    WHERE Salary > 50000;

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE Salary > 50000;
Empty set (0.01 sec)
```

14. Write a query to increase the salary of all employees in DeptID = 10 by 10%.
    UPDATE Employee
    SET Salary = Salary * 1.10
    WHERE DeptID = 10;
    SELECT EmpID, EmpName, Salary, DeptID
    FROM Employee
    WHERE DeptID = 10;

```
mysql> UPDATE Employee
    -> SET Salary = Salary * 1.10
    -> WHERE DeptID = 10;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT EmpID, EmpName, Salary, DeptID
    -> FROM Employee
    -> WHERE DeptID = 10;
Empty set (0.00 sec)
```

15. Write a query to count the total number of employees in each department.
    SELECT DeptID, COUNT(*) AS TotalEmployees
    FROM Employee
    GROUP BY DeptID;

```
mysql> SELECT DeptID, COUNT(*) AS TotalEmployees
    -> FROM Employee
    -> GROUP BY DeptID;
Empty set (0.01 sec)
```

16.Write a query to list departments that have no employees. & create a view for those salespeople who belong to the city of Mumbai.

```
SELECT d.DeptID, d.DeptName
FROM Department d
LEFT JOIN Employee e
ON d.DeptID = e.DeptID
WHERE e.EmpID IS NULL;
CREATE VIEW Mumbai_Salespeople AS
SELECT e.EmpID, e.EmpName, e.Salary, d.City
FROM Employee e
JOIN Department d
ON e.DeptID = d.DeptID
WHERE d.DeptName = 'Sales'
AND d.City = 'Mumbai';
SELECT * FROM Mumbai_Salespeople;
```

```
mysql> SELECT d.DeptID, d.DeptName
    -> FROM Department d
    -> LEFT JOIN Employee e
    -> ON d.DeptID = e.DeptID
    -> WHERE e.EmpID IS NULL;
+--------+----------+
| DeptID | DeptName |
+--------+----------+
|     10 | IT       |
|     20 | HR       |
|     30 | Sales    |
+--------+----------+
3 rows in set (0.01 sec)

mysql> CREATE VIEW Mumbai_Salespeople AS
    -> SELECT e.EmpID, e.EmpName, e.Salary, d.City
    -> FROM Employee e
    -> JOIN Department d
    -> ON e.DeptID = d.DeptID
    -> WHERE d.DeptName = 'Sales'
    -> AND d.City = 'Mumbai';
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Mumbai_Salespeople;
Empty set (0.00 sec)
```

17.Write a query to display the highest salary among all employees.

```
SELECT MAX(Salary) AS HighestSalary
FROM Employee;
```

```
mysql> SELECT MAX(Salary) AS HighestSalary
    -> FROM Employee;
+---------------+
| HighestSalary |
+---------------+
|          NULL |
+---------------+
1 row in set (0.00 sec)
```

18.Write a query to display the 2nd highest salary among all employees.

```
SELECT MAX(Salary) AS SecondHighestSalary
FROM Employee
WHERE Salary < (
  SELECT MAX(Salary)
  FROM Employee
);
```

```
mysql> SELECT MAX(Salary) AS SecondHighestSalary
    -> FROM Employee
    -> WHERE Salary < (
    ->     SELECT MAX(Salary)
    ->     FROM Employee
    -> );
+---------------------+
| SecondHighestSalary |
+---------------------+
|                NULL |
+---------------------+
1 row in set (0.04 sec)
```

19.Write a query to show all employees ordered by JoiningDate in descending order.

```
SELECT *
FROM Employee
ORDER BY JoiningDate DESC;
```

```
mysql> SELECT *
    -> FROM Employee
    -> ORDER BY JoiningDate DESC;
Empty set (0.00 sec)
```

20.Write a query to rename the DeptName column to DepartmentName.

ALTER TABLE Department
RENAME COLUMN DeptName TO DepartmentName;
DESC Department;

```
mysql> ALTER TABLE Department
    -> RENAME COLUMN DeptName TO DepartmentName;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Department;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| DeptID         | int         | NO   | PRI | NULL    |       |
| DepartmentName | varchar(50) | YES  |     | NULL    |       |
| City           | varchar(50) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
3 rows in set (0.01 sec)
```